

Vishnu Datta

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EDUCATION

Indian Institute of Information Technology, Sri City

Bachelors of Technology in Computer Science and Engineering

Aug. 2023 – May 2027

Sri City, India

EXPERIENCE

Soundverse AI

Apr 2026 – Present

Software Development Engineer Intern

Remote

- Architected a scalable transcription and speaker diarization API, extending **Whisper** processing from **25MB to 500MB+** audio inputs using silence-aware chunking, downsampling, and audio merging pipelines
- Architected end-to-end full-stack media processing features, developing high-performance **FFmpeg**- and **Pillow**-powered file conversion APIs with advanced transcoding controls alongside AI Stem Separation workflows
- Overhauled the **Admin Dashboard** with internal monitoring and operational tooling, including referral fraud detection to prevent **bot-driven commission abuse** alongside user activity tracking and token oversight
- Implemented secure **Apple Sign-In** authentication for Android using backend callback APIs, alongside subscription lifecycle management and authentication control systems
- Streamlined internal support operations by architecting **webhook-based** ticket routing automations that intelligently categorize and dispatch **Zammad** email alerts to targeted **Slack** channels for faster support resolution
- Collaborated across agile development lifecycles using Git, issue tracking, PR workflows, code reviews, bug triaging, and staging/production deployments

PROJECTS

CodeRep | *Next.js, TypeScript, PostgreSQL, Prisma, Clerk, Chrome Extension, Tailwind CSS* | GitHub 🐙 | [Live Demo](#)

- Engineered a full-stack web application with a **Chrome Extension (Manifest V3)** implementing spaced repetition algorithms for LeetCode revision, automating review scheduling for **20+ daily active users**.
- Designed RESTful APIs using **Next.js App Router** and **Prisma ORM**, implementing dual authentication (**Clerk OAuth + token-based auth**) for reliable cross-platform synchronization.
- Developed a **MutationObserver-powered** content script to detect LeetCode submissions in real time, achieving **99% detection accuracy** through intelligent submission-state filtering.
- Built an adaptive spaced-repetition scheduling system using **exponential backoff multipliers (1.2x–4x)** to dynamically reduce unnecessary review frequency while reinforcing long-term problem retention.

FluenSee | *React, Node.js, MongoDB, Google Gemini API, Socket.IO, Stream SDK, JWT* | GitHub 🐙 | [Live Demo](#)

- Developed an AI-powered language tutor using **Google Gemini 2.5 Flash**, enabling real-time conversational practice with instant grammar correction through interactive role-play based learning sessions
- Integrated high-quality **1v1 video calling** using the **Stream Video SDK** and **Socket.IO**, enabling seamless peer-to-peer language exchange with synchronized real-time communication
- Built a scalable full-stack architecture using **React + Vite, Node.js**, and **MongoDB**, implementing secure authentication with **JWT** and HTTP-only cookie-based session handling

AlgoClash (In Progress) | *React, Node.js, Socket.IO, MongoDB, Piston API* | GitHub 🐙

- Developing a real-time **1v1 competitive coding platform** using **Socket.IO**, enabling synchronized multiplayer coding battles with isolated match state management and low-latency event synchronization
- Designed an automated tournament system generating scalable bracket structures for **32+ participants**, supporting dynamic round advancement and real-time match progression
- Built a secure remote code execution pipeline using the **Piston API**, supporting compilation and execution workflows across **8 programming languages** with standardized runtime handling

TECHNICAL SKILLS

Languages: Java, JavaScript (ES6+), TypeScript, Python, C, SQL, HTML, CSS

Frontend: React.js, Next.js, Tailwind CSS, Shadcn UI, Vite

Backend & APIs: Node.js, Express.js, FastAPI, REST APIs, WebSockets, Socket.IO, Prisma ORM, JWT Authentication, OAuth, Clerk Auth, API Integration, Webhooks, Async Programming, Chrome Extension APIs

Databases & Real-Time Systems: PostgreSQL, MongoDB, Redis, Supabase, Firebase

AI, Tools & Platforms: OpenAI Whisper API, Google Gemini API, Git, GitHub, Docker, Postman, Azure, Stream